

LG-ProXAI: Local–Random–Global Perturbation Diagnostics for Parameter-Efficient Speech Emotion Recognition

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Abstract—This paper investigates speech emotion recognition (SER) with Voxtral-Mini-3B adapted by Weight-Decomposed Low-Rank Adaptation (DoRA). The model is fine-tuned on the English subset of the Emotional Speech Database (ESD), evaluated in-domain on ESD, and tested cross-corpus on IEMOCAP. The primary contribution is LG-ProXAI, a post-hoc Local–Random–Global diagnostic protocol for measuring perturbation-based sensitivity in Voxtral-based SER. LG-ProXAI compares selected temporal regions with same-duration random controls and utterance-level prosody perturbations under a fixed forced-choice label-scoring rule. Audio-only DoRA reaches 82.43% macro-F1 on ESD and 62.54% on IEMOCAP. On 1,000 predicted-class-balanced ESD predictions, mean forced-choice score drops are 0.058 for random regions, 0.187 for the highest-ranked local region, 0.228 for the combined top two regions, and 0.486 for the maximum observed global perturbation effect among eight tested utterance-level transformations. A parallel IEMOCAP diagnostic run shows the same qualitative pattern, with mixed local–global sensitivity as the dominant diagnostic pattern. The evidence is interpreted as controlled post-hoc input–output sensitivity, not as causal evidence of the model’s internal decision mechanism.

Project repository: <https://github.com/Explainable-team/explainable-project>

Index Terms—speech emotion recognition, audio-language model, DoRA, explainable AI, perturbation sensitivity, cross-corpus evaluation

I. INTRODUCTION

Speech emotion recognition (SER) enables affect-aware interfaces and assistive systems, but responsible deployment requires more than aggregate predictive accuracy. Emotion labels depend on prosodic, lexical, speaker, and recording factors, and SER models often degrade across corpora and speakers [4]. Automatic emotion recognition also raises privacy, misuse, and over-interpretation risks; therefore, model evaluation should include evidence-aware diagnostics rather than prediction scores alone [15].

Audio-language models such as Voxtral can process speech and text to generate emotion labels [1]. Because full fine-tuning is computationally expensive, this work uses DoRA, a parameter-efficient extension of LoRA, to adapt Voxtral-Mini-3B for SER [2], [3]. However, adaptation does not explain whether a prediction depends on localized acoustic evidence, generic perturbation fragility, or distributed utterance-level prosody.

This paper introduces LG-ProXAI, a controlled post-hoc diagnostic protocol for Voxtral-based SER. The objective is not to claim state-of-the-art emotion recognition, but to audit

whether perturbation-based evidence separates local temporal sensitivity from random-control sensitivity and whole-utterance prosodic dependence.

II. RELATED WORK

Model-agnostic methods such as LIME and SHAP estimate local feature contributions [7], [8], while Integrated Gradients provides an axiomatic attribution framework [9]. Perturbation methods directly modify the input and measure output changes [10]. Their evaluation has motivated parameter-randomization checks [11], remove-and-retrain benchmarks [18], infidelity and sensitivity metrics [17], and toolkits such as Quantus [20]. These studies show that explanation quality is multidimensional and cannot be inferred from visualization alone [19].

In audio, audioLIME produces listenable explanations by perturbing source-separated components [13]. SpeechXAI combines word-level audio segments with counterfactual changes in pitch, speaking rate, and noise [14]. Comprehensiveness and sufficiency, popularized by ERASER, test whether selected evidence is necessary and sufficient under a chosen scoring rule [12]. Prior speech-XAI work mainly targets encoder classifiers, word-aligned evidence, or selected examples. Generative audio-language models require a stricter comparison policy: original and perturbed inputs must be compared with the same fixed label-scoring rule rather than free-form text generation.

III. RESEARCH GAP

Three gaps motivate this study. First, parameter-efficient adaptation methods such as LoRA and DoRA reduce the cost of specializing large models, but SER studies using large audio or audio-language models rarely analyze the acoustic sensitivity of the adapted output [2]–[4]. Second, local speech explanations are often reported without same-duration random controls and utterance-level prosody baselines, which can confuse localization with generic perturbation fragility [10], [16], [17]. Third, qualitative sample-level explanations are seldom connected to class-balanced statistics, error fragility, and cross-corpus behavior, although rigorous XAI evaluation requires more than visually plausible explanations [11], [18], [20].

The study addresses two research questions: *RQ1: Which temporal regions and acoustic transformations produce the largest changes in the predicted-label score?* *RQ2: Are these effects stronger than same-duration random controls and*

distinguishable from whole-utterance prosody sensitivity? The contribution is a post-hoc diagnostic workflow that combines local temporal attenuation, approximate random controls, global prosody perturbations, contrastive margin changes, and class-level error analysis. It is not presented as a causal explanation method or as a theory of emotional expression.

IV. METHODOLOGY AND IMPLEMENTATION

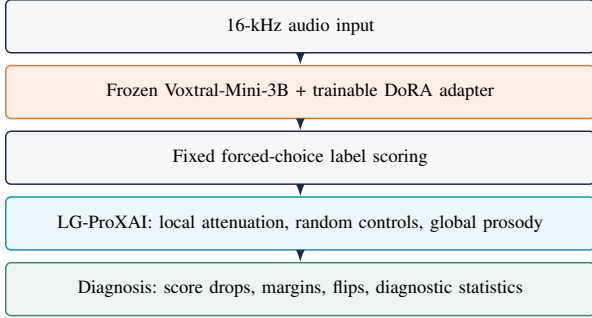


Fig. 1. Compact explainability-centered workflow. Original and perturbed audio variants are compared under the same forced-choice label-scoring rule.

A. Experimental Setup

The workflow contains data auditing, zero-shot sanity checking, DoRA adaptation, in-domain/cross-corpus evaluation, and LG-ProXAI diagnosis (Fig. 1). ESD provides controlled emotional speech, and its English subset is used for adaptation and in-domain analysis [5]. IEMOCAP provides natural dyadic dialogue and serves as a cross-corpus domain-shift test [6]. The four-class setup uses Angry, Happy, Sad, and Neutral. The ESD Surprise class is excluded from the adapted setup, and IEMOCAP excitement is merged with happiness. The report emphasizes the audio-only condition so that the explanation analysis refers to acoustic evidence rather than transcript-based information.

DoRA uses rank $r = 16$, scale $\alpha = 32$, and dropout 0.1. The adapter is applied to the self-attention projection modules $\{q, k, v, o\}$. Training uses three epochs, learning rate 10^{-4} , cosine decay, mixed precision, gradient accumulation over 16 steps, and early stopping based on validation loss.

B. Forced-Choice Label Scoring

For the XAI stage, original and perturbed audio variants are compared with a fixed forced-choice label-token scoring rule. For each class verbalizer y_c , the length-normalized log-likelihood is

$$s_c(x) = \frac{1}{T_c} \sum_{t=1}^{T_c} \log p(y_{c,t} | x, y_{c,<t}), \quad (1)$$

where $p(\cdot)$ is produced by the multimodal autoregressive decoder for input audio x . Candidate labels are normalized over the fixed label set:

$$q_c(x) = \frac{\exp(s_c(x))}{\sum_j \exp(s_j(x))}. \quad (2)$$

Here, $q_c(x)$ is a normalized forced-choice label score, not a calibrated emotion probability. For the predicted class $\hat{c}$, the perturbation effect is

$$\Delta_{\hat{c}}(x, \tilde{x}) = q_{\hat{c}}(x) - q_{\hat{c}}(\tilde{x}), \quad (3)$$

where $\tilde{x}$ is the perturbed audio. Larger positive values indicate a stronger decrease in the predicted-label score under the tested intervention. Classification metrics are obtained from the evaluation pipeline; XAI metrics are computed by re-scoring original and perturbed variants with this fixed rule.

C. Local, Random-Control, and Global Interventions

Speech-active candidate regions are divided into 0.5-s temporal windows. Each candidate is attenuated by multiplying its waveform samples by a fixed factor $\alpha_m = 0.2$ while leaving the rest of the utterance unchanged. Perturbed variants are ranked by log-margin drop, probability-margin drop, and predicted-label score drop. Top-1, Top-2, and their union represent selected local evidence.

Random controls use regions of identical duration sampled from the utterance while avoiding the selected top evidence region. They estimate non-specific perturbation sensitivity, but they are approximate controls rather than perfectly matched counterfactuals because they are not fully matched by energy, voicing, temporal position, or speech activity. Global interventions modify the whole waveform through eight fixed transformations: energy down/up, pitch down/up, time faster/slower, and spectral darken/brighten. The reported global statistic is the maximum observed global perturbation effect, defined as the largest predicted-label score drop among the eight tested utterance-level transformations. It should therefore be interpreted as the strongest observed global sensitivity under the tested perturbation set, not as an average global effect or as an exhaustive maximum over all possible prosodic changes.

TABLE I
LG-PROXAI COMPONENTS AND DIAGNOSTIC OUTPUTS.

Component	Diagnostic question	Reported quantity
Local attenuation	Which regions affect the predicted score?	Δq , margin change, ranks
Random controls	Is local evidence stronger than random perturbation?	Local-random gap, empirical z
Global prosody	How sensitive is the output to utterance-level change?	Maximum observed drop, flip rate
Deletion	Does removing evidence reduce the score?	Comprehensiveness curve
Keep-only	Does retained evidence preserve the score?	Sufficiency curve
Class statistics	Do patterns vary across classes and errors?	Means, ratios, error-stratified summaries

For top- k regions $R_{1:k}$, comprehensiveness is $q_{\hat{c}}(x) - q_{\hat{c}}(x \setminus R_{1:k})$. Sufficiency is $q_{\hat{c}}(x) - q_{\hat{c}}(R_{1:k})$; values close to zero indicate that retained evidence preserves the predicted-label score under the tested scorer [12]. Local specificity is summarized as

$$z = \frac{\Delta_{\text{local}} - \mu_{\text{random}}}{\sigma_{\text{random}}}.$$

Ratios to global effects are reported only when the denominator is non-negligible and the local and global effects have the same sign.

The 1,000-sample ESD explanation run is balanced by predicted class, with 250 samples per predicted class. It contains 801 correct and 199 incorrect predictions and covers 30,978 perturbed variants. A parallel IEMOCAP diagnostic run uses the same predicted-class-balanced design, with 250 samples per predicted class, 800 correct predictions, 200 incorrect predictions, and 30,846 perturbed variants. Detailed deletion, keep-only, localized counterfactual, and human-readable reports are restricted to selected 20-sample qualitative analyses.

V. RESULTS AND ANALYSIS

A. Model Performance

TABLE II
AUDIO-ONLY DORA PERFORMANCE WITH $r = 16$ AND $\alpha = 32$.

Metric	ESD	IEMOCAP
Accuracy (%)	82.64	61.75
Balanced accuracy (%)	82.64	65.02
Macro-F1 (%)	82.43	62.54
Matthews correlation coefficient	0.7729	0.4990
Cohen's κ	0.7686	0.4894

Table II shows stronger in-domain performance on ESD than cross-corpus performance on IEMOCAP. This gap motivates explanation analysis because aggregate accuracy alone cannot determine whether errors arise from unstable local evidence, non-specific perturbation fragility, or sensitivity to distributed prosodic variation.

B. Explanation Diagnostics

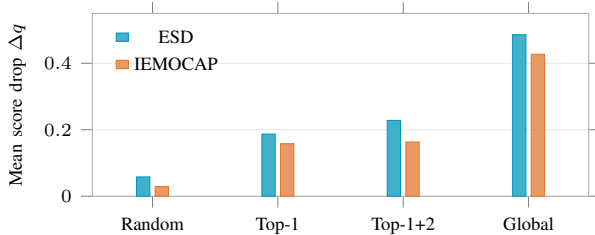


Fig. 2. Cross-corpus LG-ProXAI sensitivity comparison. Values are mean forced-choice score drops on 1,000 predicted-class-balanced samples per dataset. “Global” denotes the maximum observed score drop among eight tested whole-utterance transformations. Exact values are reported in Table III.

TABLE III
COMPACT CROSS-CORPUS LG-PROXAI SUMMARY. SCORE DROPS ARE MEAN FORCED-CHOICE SCORE DROPS; RATIOS SUMMARIZE RELATIVE SENSITIVITY.

Diagnostic quantity	ESD	IEMOCAP
Random control drop	0.058	0.029
Top-1 local drop	0.187	0.158
Top-1 + Top-2 drop	0.228	0.163
Maximum observed global drop	0.486	0.427
Top-1 / random ratio	3.22	5.40
Maximum global / Top-1 ratio	2.60	2.71
Dominant diagnostic pattern	Mixed local–global	Mixed local–global
Dominant global sensitivity type	Rate (50.5%)	Rate (50.0%)
Incorrect/correct local ratio	1.62	1.47

Figure 2 and Table III show a consistent cross-corpus pattern. In both datasets, selected local regions produce substantially larger forced-choice score drops than same-duration random controls. This indicates that the top-ranked temporal regions contain evidence beyond non-specific perturbation sensitivity under the tested perturbation protocol. However, the maximum observed global perturbation effect is larger than the local effects in both corpora, showing that the adapted model is not explained by isolated local evidence alone.

The ratios in Table III make this behavior explicit. Top-1 local evidence is 3.22 times larger than the random-control baseline on ESD and 5.40 times larger on IEMOCAP. At the same time, the maximum observed global perturbation effect is 2.60 times larger than Top-1 local evidence on ESD and 2.71 times larger on IEMOCAP. Therefore, the dominant diagnostic pattern is mixed local–global sensitivity: local temporal evidence matters, but utterance-level prosody remains the larger measured source of perturbation sensitivity under the tested transformations.

Mixed local–global sensitivity is the dominant diagnostic pattern in both datasets: 40.5% on ESD and 46.3% on IEMOCAP. Speaking-rate sensitivity is also the dominant global sensitivity type in both corpora, accounting for 50.5% of ESD samples and 50.0% of IEMOCAP samples. This suggests that the adapted model relies strongly on distributed timing-related cues, which are plausible acoustic correlates of emotion but also potential sources of cross-corpus fragility.

Incorrect predictions show higher perturbation sensitivity than correct predictions. The incorrect/correct local-sensitivity ratio is 1.62 on ESD and 1.47 on IEMOCAP. This supports the interpretation that errors are associated with more fragile input–output behavior under controlled perturbations.

A correct Angry example illustrates the qualitative report format. The original predicted-label score is 0.949. Attenuating the highest-ranked region (0.29–0.79 s) reduces the Angry score by 0.611 and the contrastive log-margin against Happy by 3.850. Slowing only this region produces the largest local sensitivity ($\Delta q = 0.819$), with $z = 3.16$ relative to random regions and a local/global ratio of 1.05. This evidence remains sample-specific perturbation sensitivity and does not prove causal determination of the emotion.

C. Limitations and Interpretation Safeguards

The analysis has five main limitations. First, $q_c(x)$ is a normalized forced-choice label score, not a calibrated emotion probability. Second, perturbations can introduce out-of-distribution audio artifacts, especially for attenuation, pitch shifting, time-scale modification, and spectral tilt. Third, random controls provide an approximate specificity baseline but are not fully matched by energy, voicing, temporal position, or speech activity. Fourth, the 1,000-sample explanation subsets are balanced by predicted class; therefore, the reported statistics describe behavior conditional on model output classes rather than the natural ground-truth distribution. Fifth, the behavior taxonomy uses predefined score-drop and margin-drop

thresholds, so its categories should be interpreted as diagnostic summaries rather than ground-truth explanation classes.

Quality checks validate output completeness and numerical sanity, not explanation faithfulness. Consequently, LG-ProXAI should be interpreted as controlled post-hoc input–output sensitivity analysis rather than causal evidence of the model’s internal decision mechanism [16], [17], [20].

VI. CONCLUSION

This work adapts Voxtral-Mini-3B for SER using DoRA and evaluates it in in-domain ESD and cross-corpus IEMOCAP settings. The performance gap between ESD and IEMOCAP confirms that SER evaluation should go beyond aggregate accuracy. LG-ProXAI complements predictive evaluation by measuring how forced-choice label scores change under local temporal attenuation, same-duration random controls, and whole-utterance prosody perturbations.

The diagnostic results show a consistent pattern across both datasets: selected local regions are more influential than random controls, but the maximum observed global perturbation effect exceeds the corresponding local effects. This indicates that the adapted model combines localized temporal evidence with distributed prosodic sensitivity. Incorrect predictions are more perturbation-sensitive than correct predictions, suggesting that fragility under controlled perturbations can support error diagnosis. Future work should add repeated seeds, stronger matched random controls, speaker-disjoint validation, calibration analysis for forced-choice scores, and direct comparisons with additional speech-XAI baselines.

REFERENCES

- [1] A. H. Liu *et al.*, “Voxtral,” *arXiv preprint arXiv:2507.13264*, 2025.
- [2] S.-Y. Liu, C.-Y. Wang, H. Yin, P. Molchanov, Y.-C. F. Wang, K.-T. Cheng, and M.-H. Chen, “DoRA: Weight-Decomposed Low-Rank Adaptation,” in *Proc. 41st Int. Conf. Mach. Learn.*, ser. PMLR, vol. 235, 2024, pp. 32100–32121.
- [3] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-Rank Adaptation of Large Language Models,” in *Proc. Int. Conf. Learn. Representations*, 2022.
- [4] S. Latif, R. Rana, S. Khalifa, R. Jurdak, J. Qadir, and B. W. Schuller, “Survey of Deep Representation Learning for Speech Emotion Recognition,” *IEEE Trans. Affect. Comput.*, vol. 14, no. 2, pp. 1634–1654, 2023.
- [5] K. Zhou, B. Sisman, R. Liu, and H. Li, “Emotional Voice Conversion: Theory, Databases and ESD,” *Speech Commun.*, vol. 137, pp. 1–18, 2022.
- [6] C. Busso *et al.*, “IEMOCAP: Interactive Emotional Dyadic Motion Capture Database,” *Lang. Resour. Eval.*, vol. 42, no. 4, pp. 335–359, 2008.
- [7] M. T. Ribeiro, S. Singh, and C. Guestrin, “Why Should I Trust You?: Explaining the Predictions of Any Classifier,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2016, pp. 1135–1144.
- [8] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017.
- [9] M. Sundararajan, A. Taly, and Q. Yan, “Axiomatic Attribution for Deep Networks,” in *Proc. 34th Int. Conf. Mach. Learn.*, ser. PMLR, vol. 70, 2017, pp. 3319–3328.
- [10] R. C. Fong and A. Vedaldi, “Interpretable Explanations of Black Boxes by Meaningful Perturbation,” in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 3449–3457.
- [11] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, “Sanity Checks for Saliency Maps,” in *Adv. Neural Inf. Process. Syst.*, vol. 31, 2018.
- [12] J. DeYoung, S. Jain, N. F. Rajani, E. Lehman, C. Xiong, R. Socher, and B. C. Wallace, “ERASER: A Benchmark to Evaluate Rationalized NLP Models,” in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics*, 2020, pp. 4443–4458.
- [13] V. Haunschmid, E. Manilow, and G. Widmer, “audioLIME: Listenable Explanations Using Source Separation,” in *Proc. 13th Int. Workshop Mach. Learn. Music*, 2020, pp. 20–24.
- [14] E. Pastor, A. Koudounas, G. Attanasio, D. Hovy, and E. Baralis, “Explaining Speech Classification Models via Word-Level Audio Segments and Paralinguistic Features,” in *Proc. 18th Conf. Eur. Chapter Assoc. Comput. Linguistics*, 2024, pp. 2221–2238.
- [15] S. M. Mohammad, “Ethics Sheet for Automatic Emotion Recognition and Sentiment Analysis,” *Comput. Linguistics*, vol. 48, no. 2, pp. 239–278, 2022.
- [16] A. Jacovi and Y. Goldberg, “Towards Faithfully Interpretable NLP Systems: How Should We Define and Evaluate Faithfulness?” in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics*, 2020, pp. 4198–4205.
- [17] C.-K. Yeh, C.-Y. Hsieh, A. Suggala, D. I. Inouye, and P. K. Ravikumar, “On the (In)fidelity and Sensitivity of Explanations,” in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.
- [18] S. Hooker, D. Erhan, P.-J. Kindermans, and B. Kim, “A Benchmark for Interpretability Methods in Deep Neural Networks,” in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.
- [19] F. Doshi-Velez and B. Kim, “Towards a Rigorous Science of Interpretable Machine Learning,” *arXiv preprint arXiv:1702.08608*, 2017.
- [20] A. Hedström, L. Weber, D. Bareeva, D. Krakowczyk, F. Motzkus, W. Samek, S. Lapuschkin, and M. M.-C. Höhne, “Quantus: An Explainable AI Toolkit for Responsible Evaluation of Neural Network Explanations and Beyond,” *J. Mach. Learn. Res.*, vol. 24, no. 34, pp. 1–11, 2023.